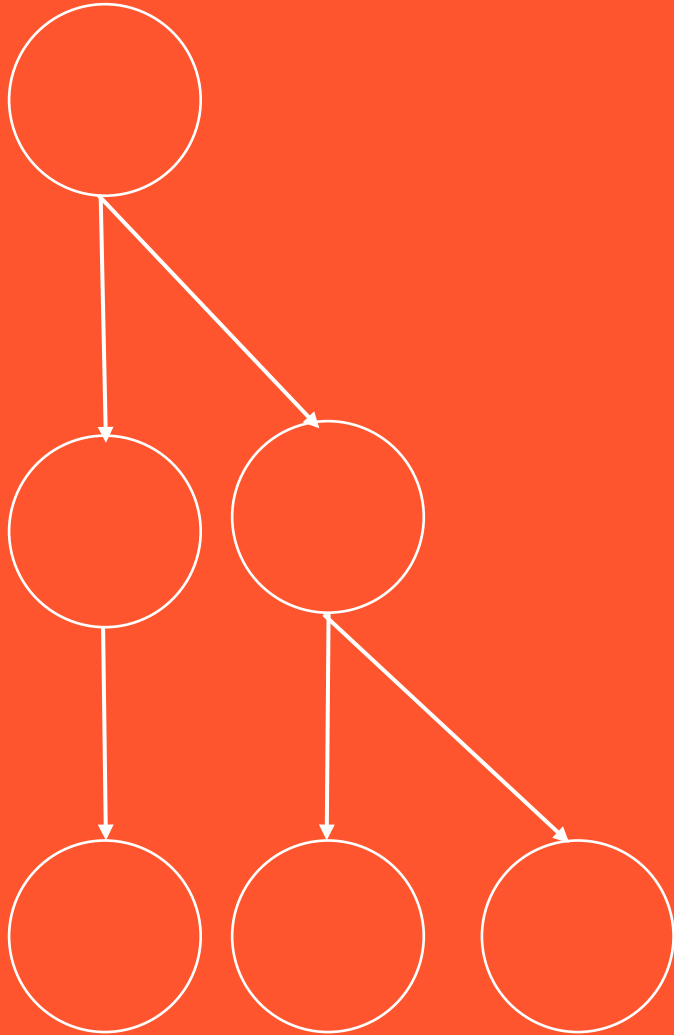


# BCOG 100

Memory, Simulation, & Generative Models

Lecture 3: Memory & Simulation in the Brain



UNIVERSITY OF  
**ILLINOIS**  
URBANA-CHAMPAIGN

## Reading Check 9-3

Password: cortex

Think about your oldest memory?  
Given what we've learned in this  
class so far, why do you think it is  
your first?

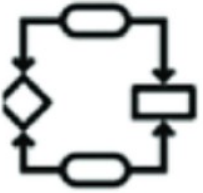


# Memory and Simulation in the Brain



## Computation

*What does the system do?*



## Algorithm

*How does the system do it?*



## Implementation

*How is the system realized?*

## Recap & Overview

### Last Time:

How prediction works.

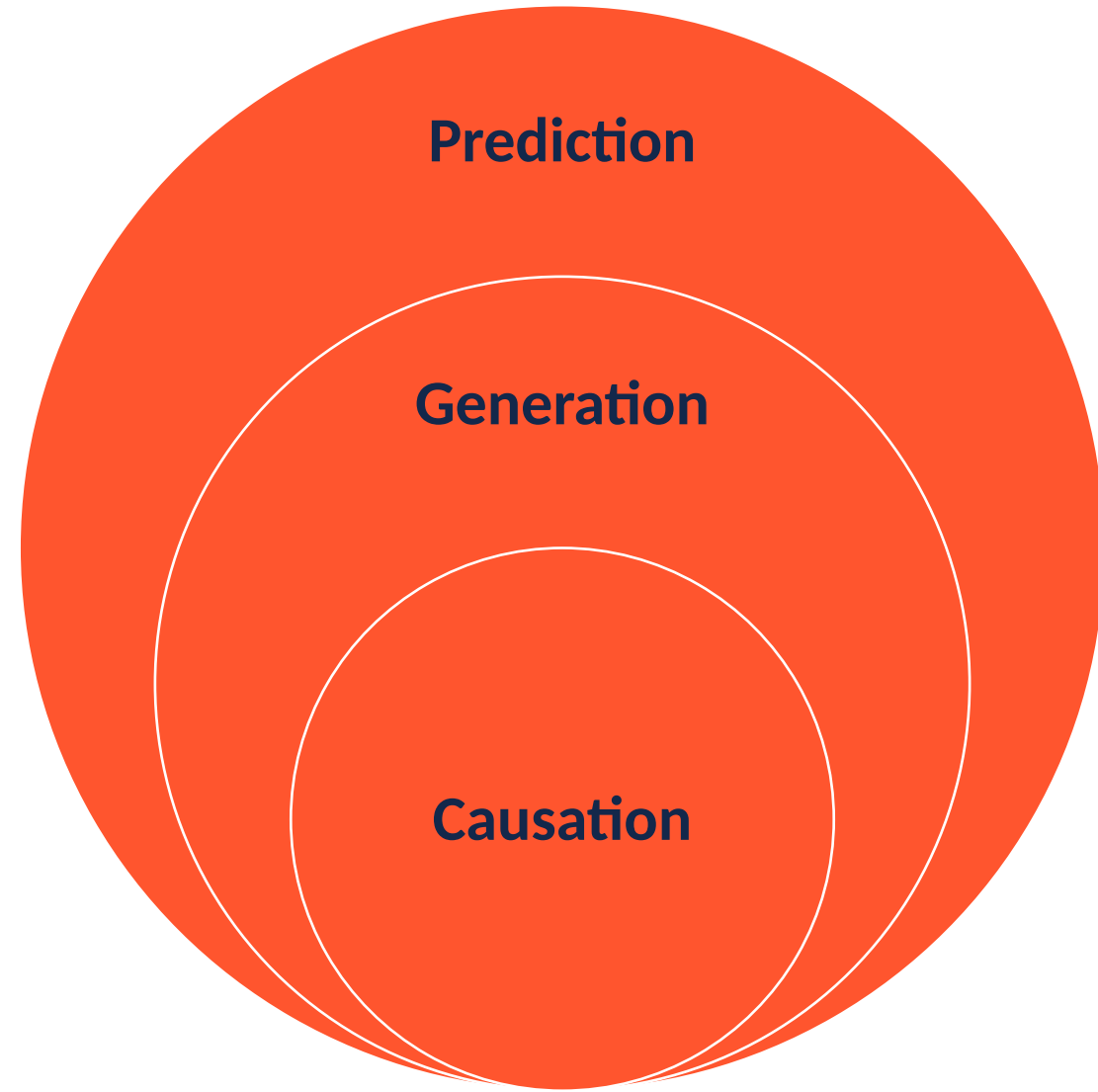
### Today:

- Where it happens and how it is implemented.
- Brains turn algorithms into anatomy.

# Memory and Simulation in the Brain

## The Brain as a Hierarchy of Predictors

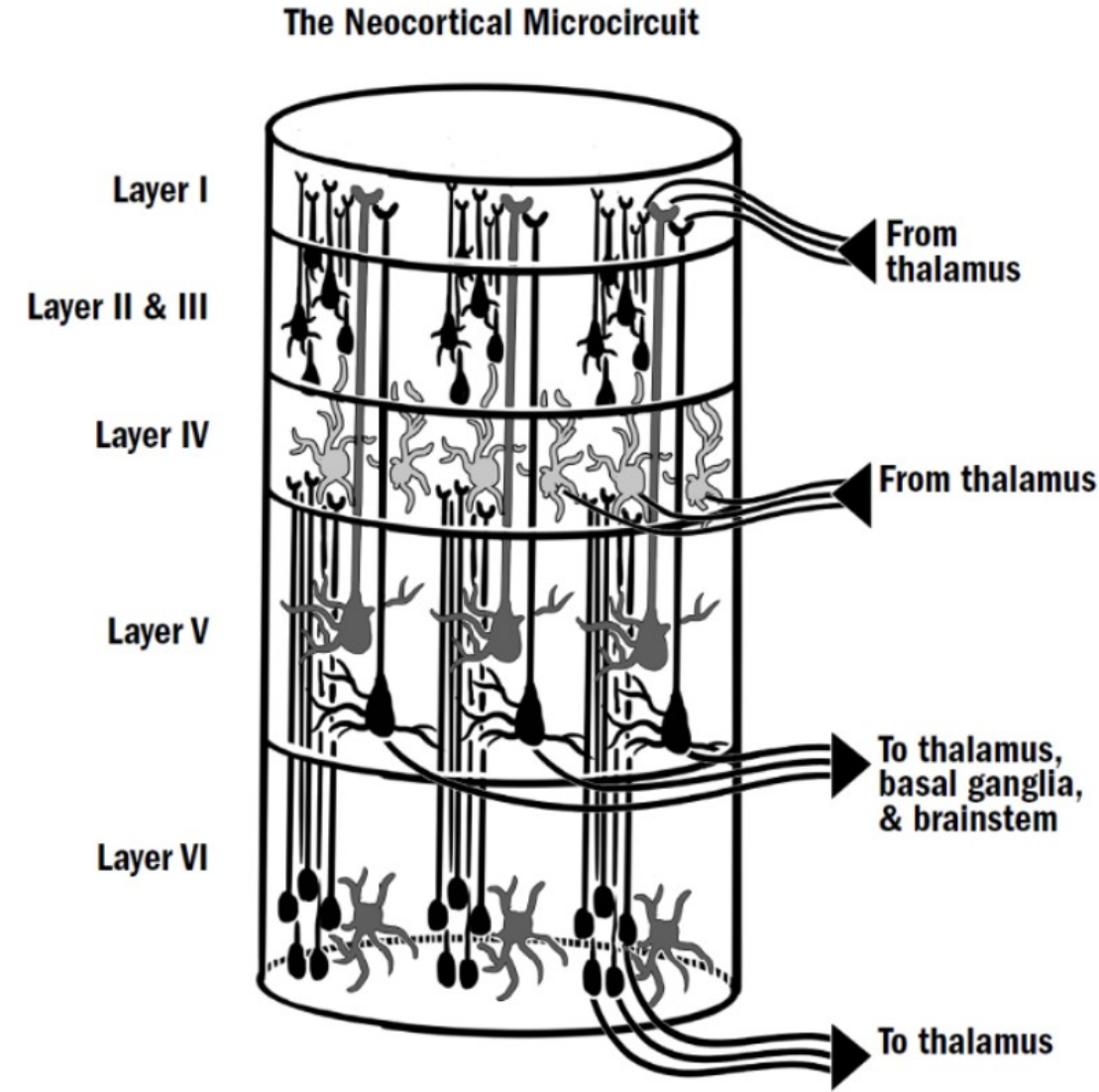
- Higher areas predict lower areas.
- Lower areas report prediction error.
- Perception is prediction passing the test.



# Memory and Simulation in the Brain

## Cortical Architecture: Layers and Loops

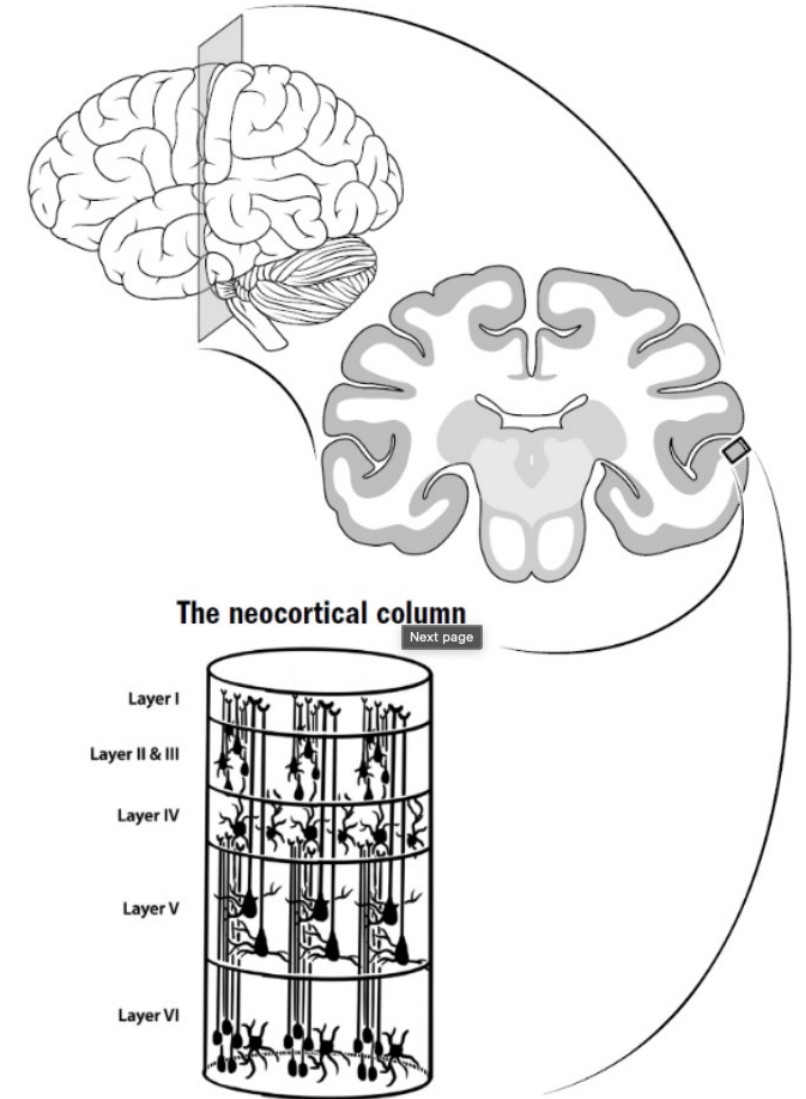
- The neocortex is a six-layer loop.
- Feedforward signals carry data up.
- Feedback signals carry predictions down.



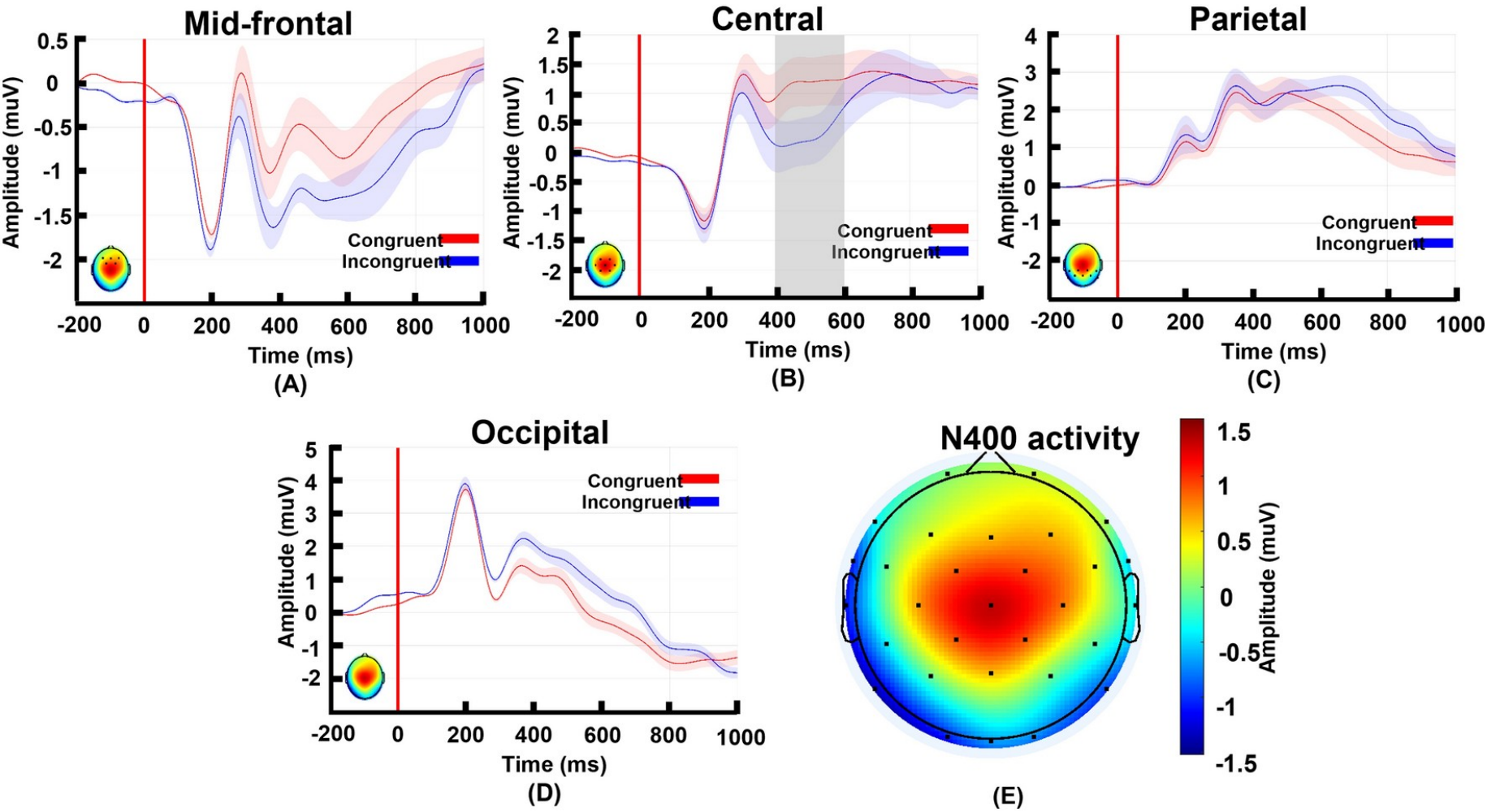
# Memory and Simulation in the Brain

## Cortical Columns as Mini-Predictors

- Each column learns local regularities.
- Together, they form a world model.
- Prediction is built from many tiny guesses.



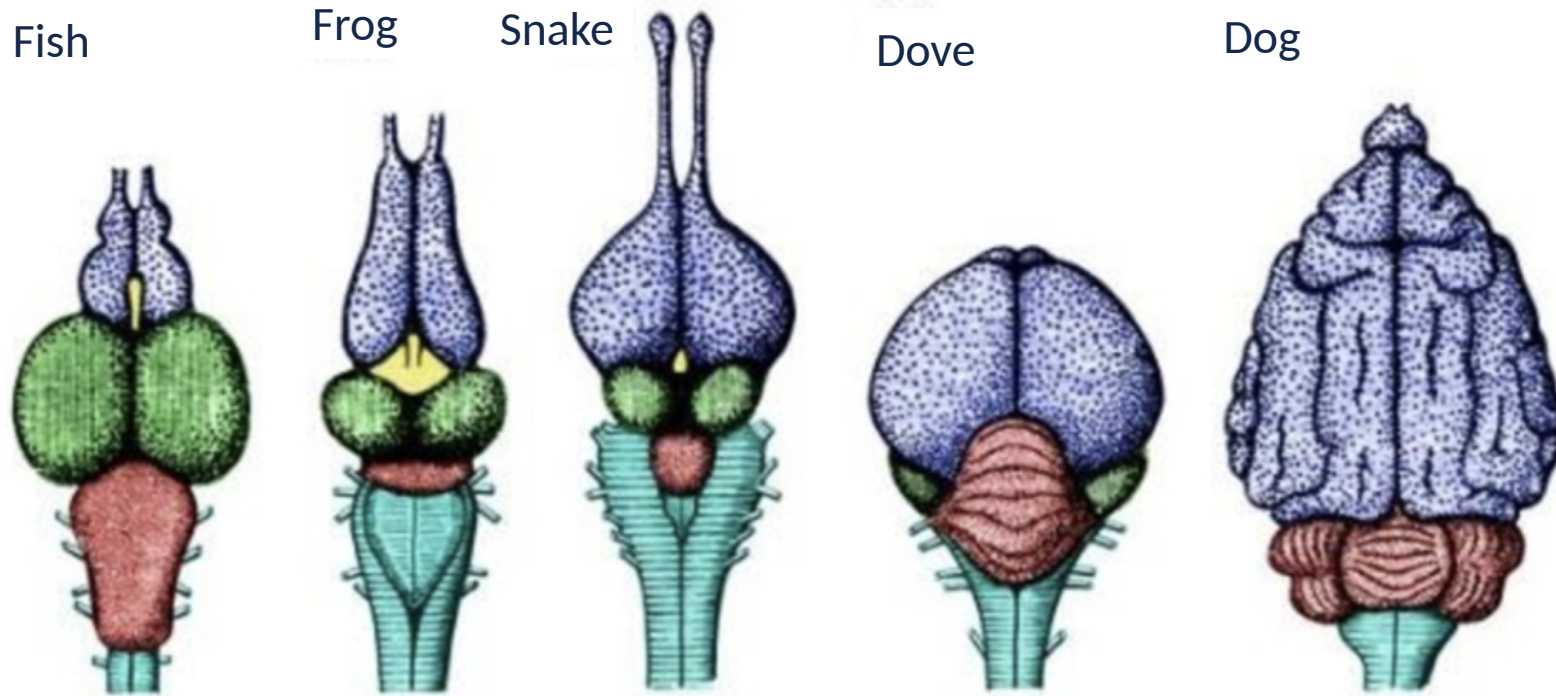
# Memory and Simulation in the Brain



## Prediction Error in Sensory Cortex

- Sensory neurons fire when predictions fail.
- Expectation dampens activity.
- Surprise lights the brain up.

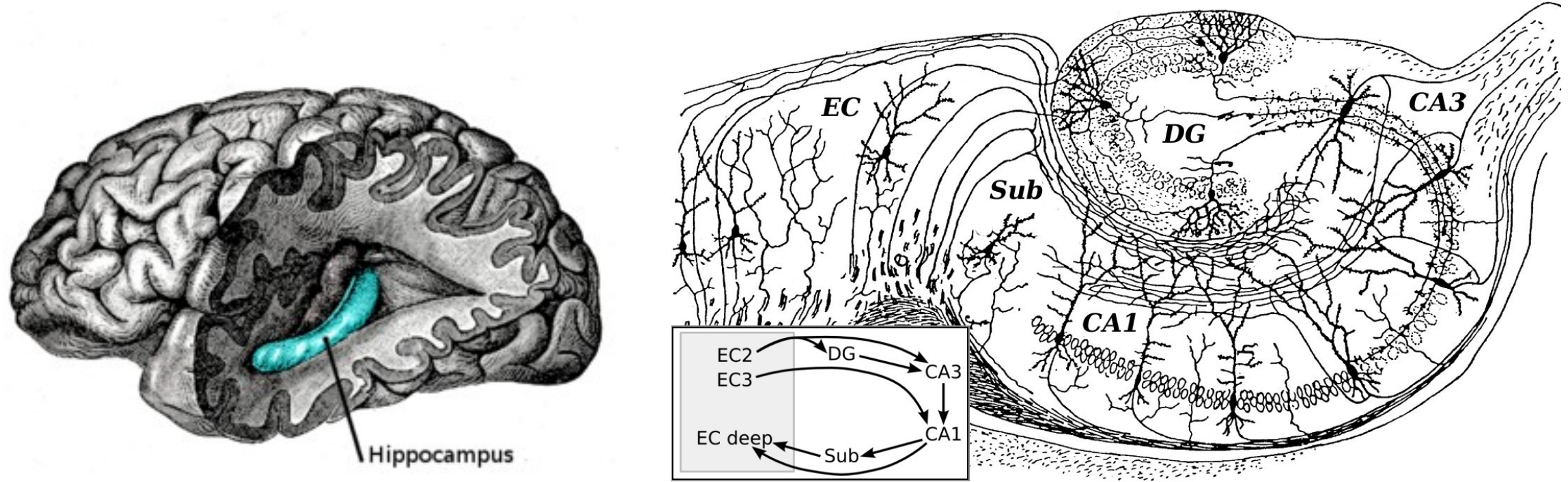
# Memory and Simulation in the Brain



## The Predictive Brain Across Species

- Prediction is older than the cortex.
- Different brains, same algorithm.
- Evolution reuses the predictive loop.

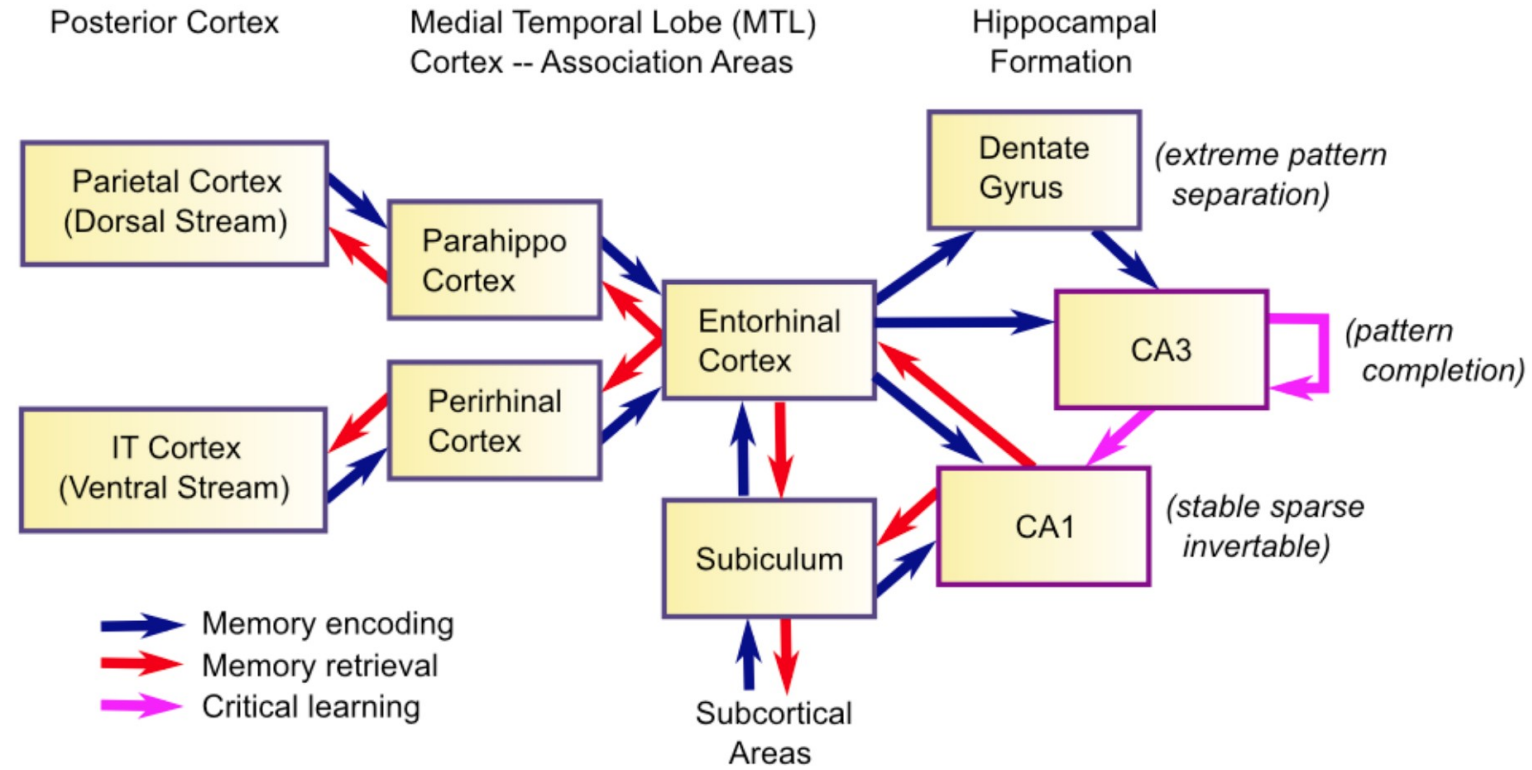
# Memory and Simulation in the Brain



## The Hippocampus: Mapping Space and Memory

- The hippocampus builds maps of experience.
- Place, grid, and head-direction cells simulate space.
- From navigation to memory: the same code.

# Memory and Simulation in the Brain



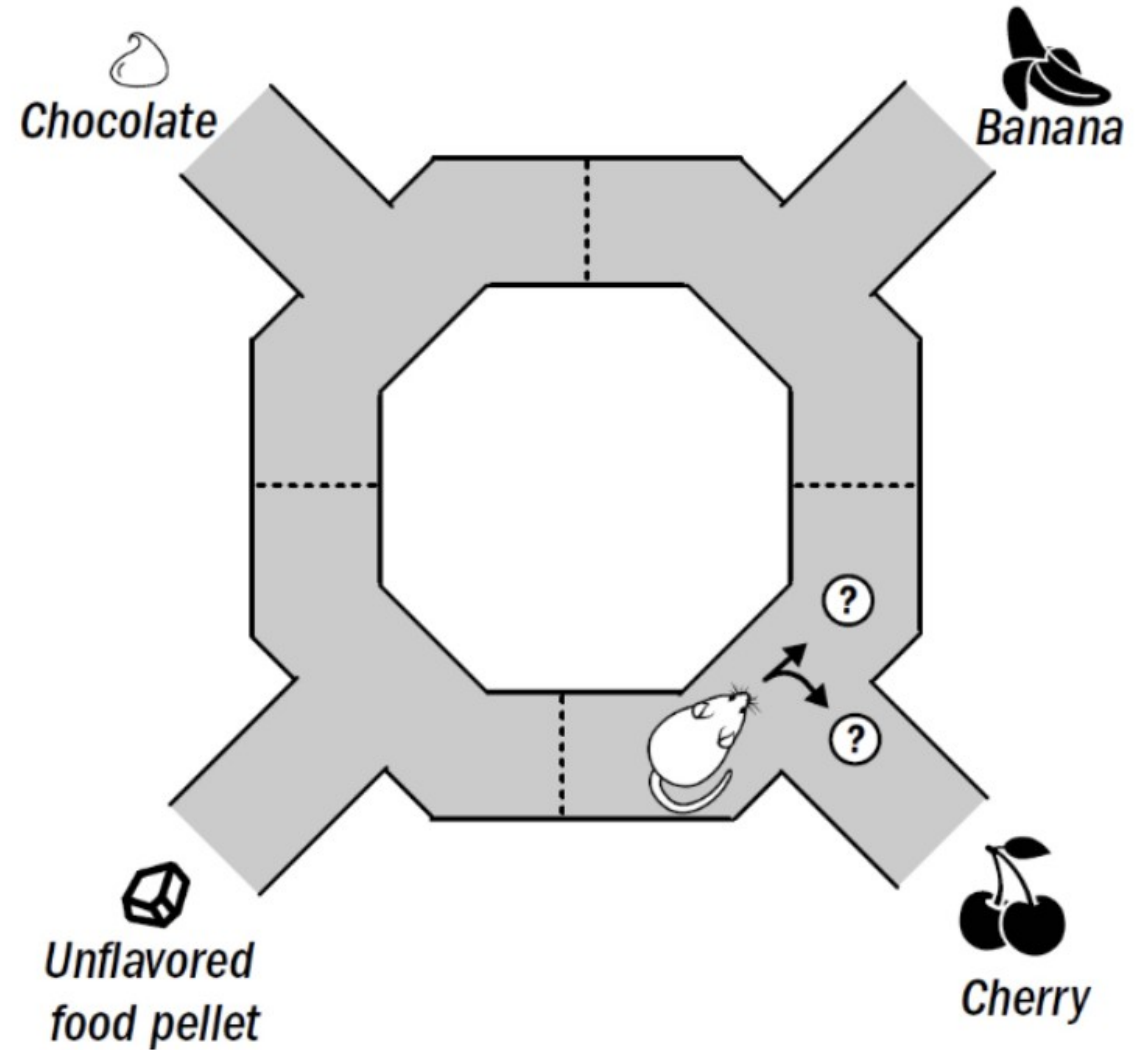
## Binding “What” and “Where”: Entorhinal Inputs to the Hippocampus

- The hippocampus builds maps of experience.
- Place, grid, and head-direction cells simulate space.
- From navigation to memory: the same code.

# Memory and Simulation in the Brain

## Replay and Preplay: The Neural Basis of Imagination

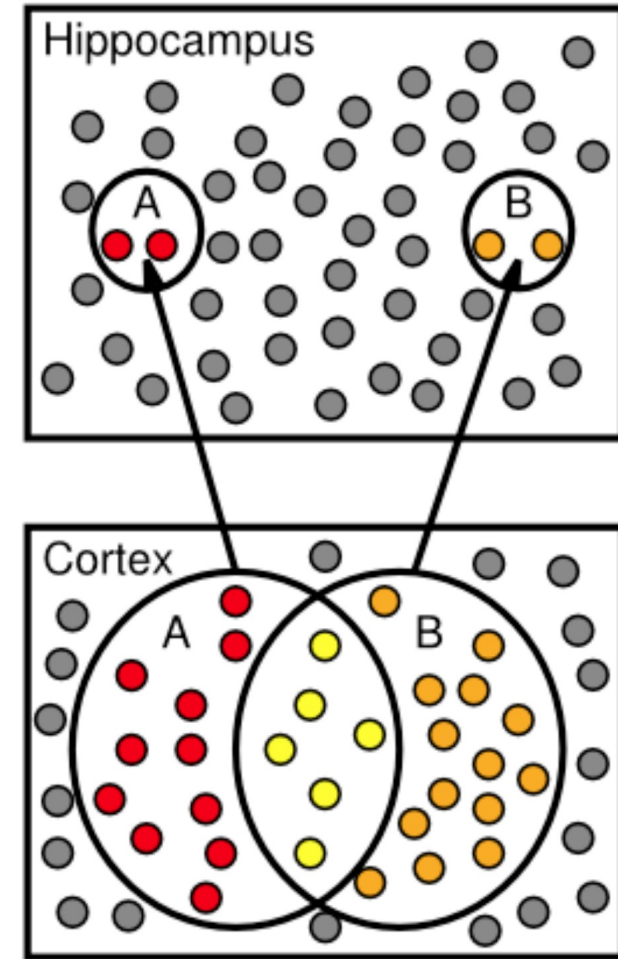
- Hippocampal sequences replay the past—and preplay the future.
- Simulation links memory to planning.
- The same circuit encodes, rehearses, and predicts.



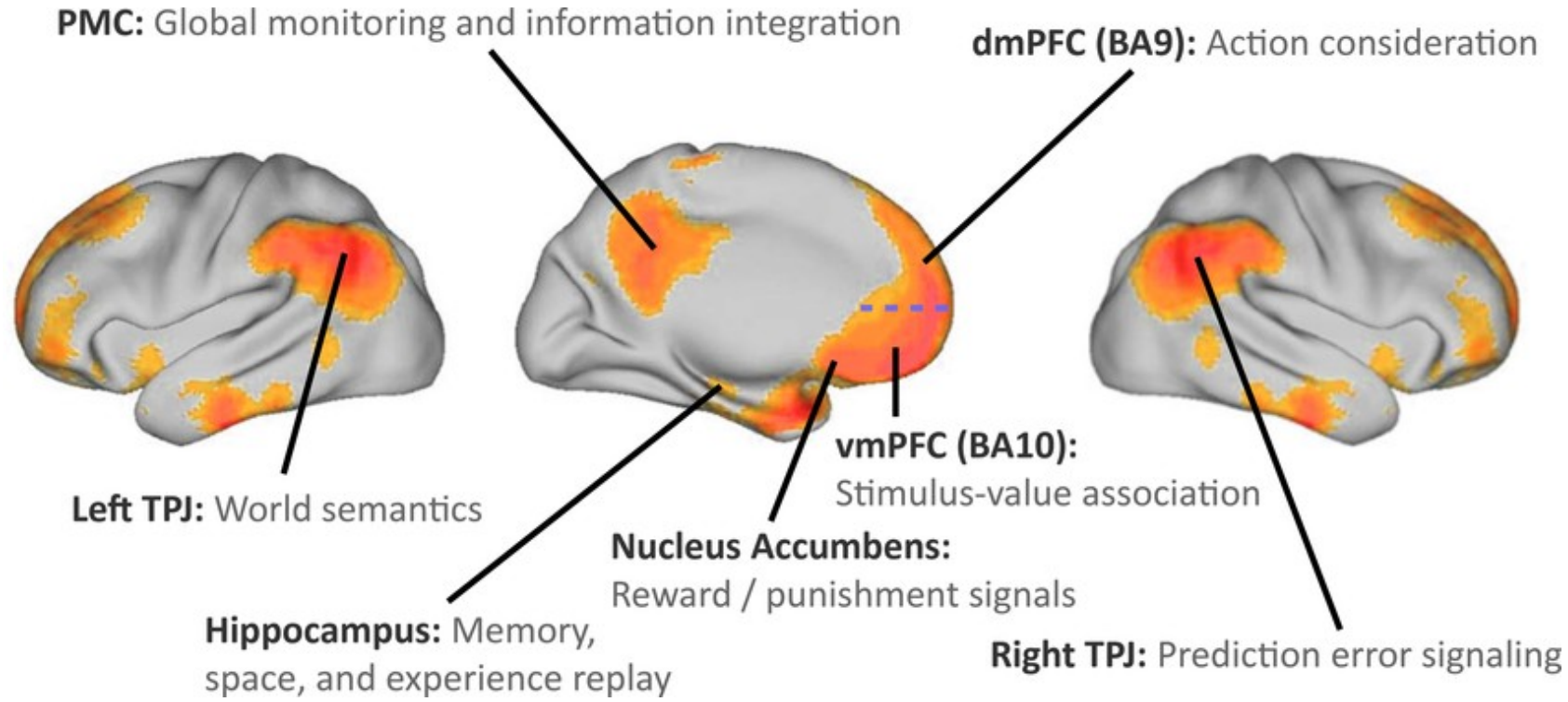
# Memory and Simulation in the Brain

## The Hippocampus–Cortex Dialogue: Memory Consolidation and Integration

- Hippocampus stores new episodes fast.
- Cortex learns slowly, through replay.
- Memory is a conversation between the two.



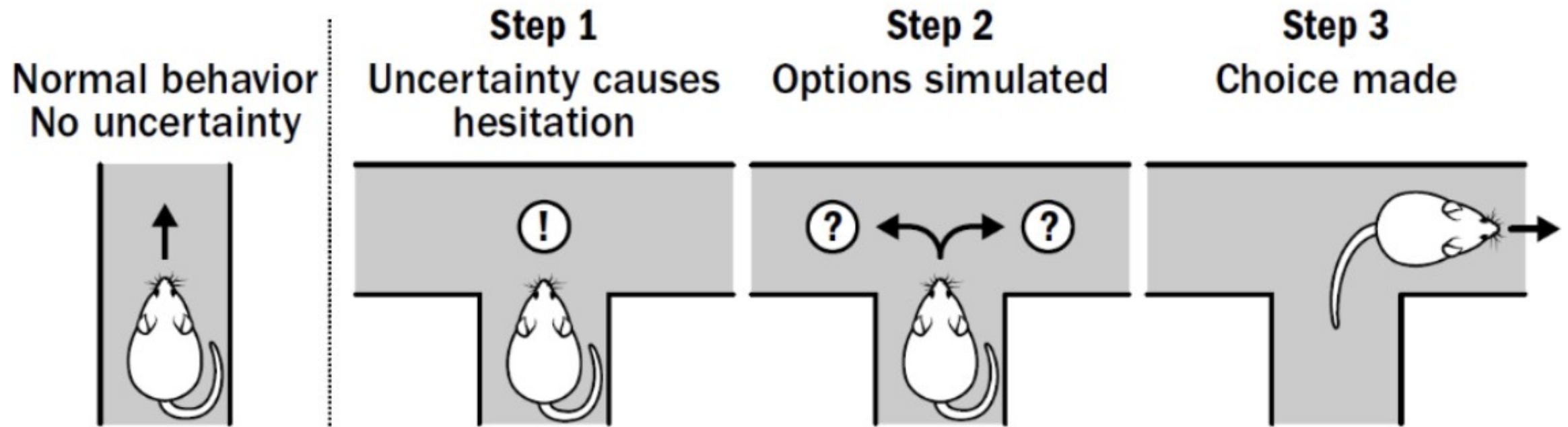
# Memory and Simulation in the Brain



## The Default-Mode Network: The Brain's Simulation Space

- When the brain is “at rest,” it’s not idle.
- The default-mode network simulates worlds, selves, and times.
- Imagination is the brain’s baseline.

# Memory and Simulation in the Brain



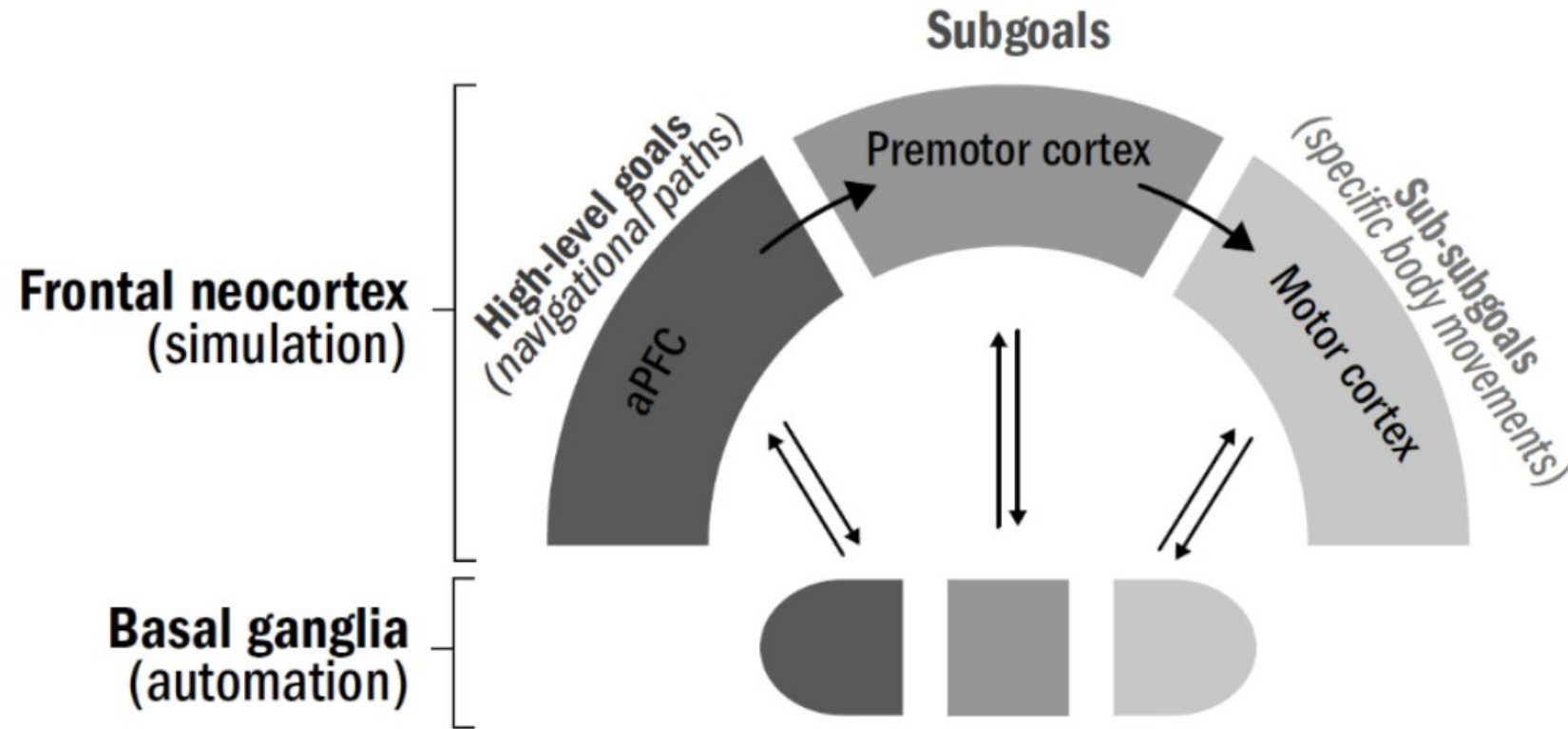
## Prefrontal Cortex: Modeling Goals and Futures

- The prefrontal cortex simulates actions before doing them.
- It builds goals, plans, and counterfactuals.
- Imagination becomes control.

# Memory and Simulation in the Brain

## Basal Ganglia: Selecting Simulated Actions

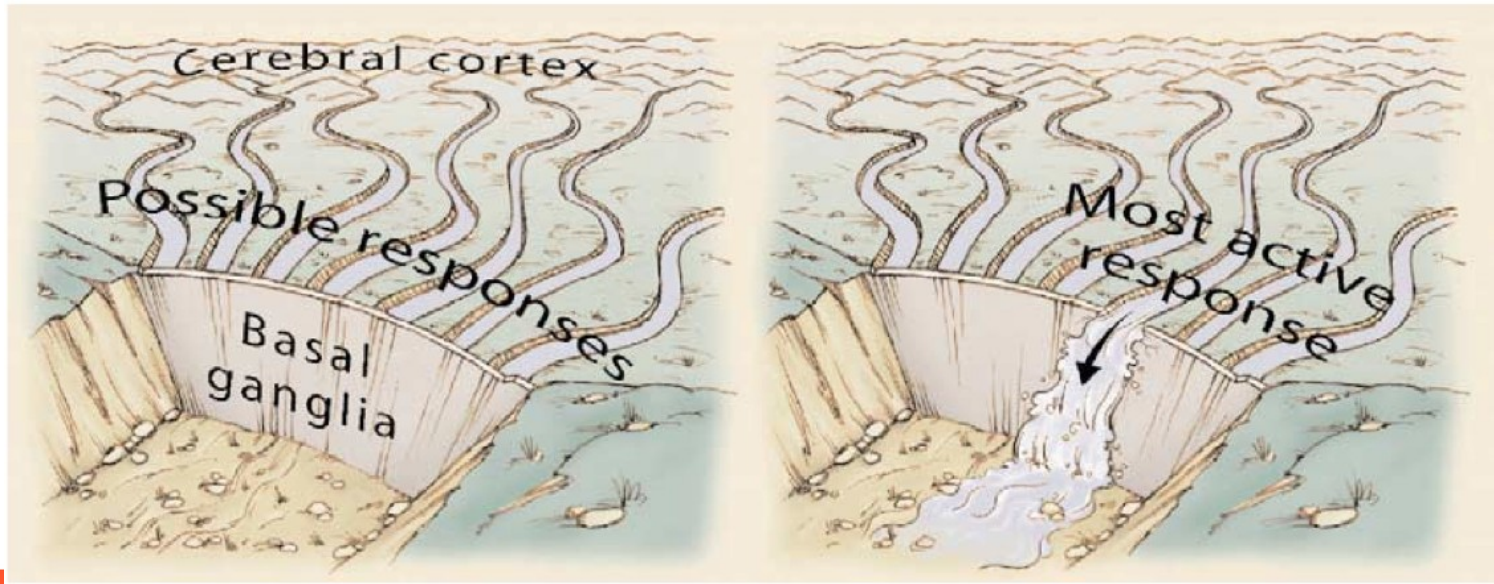
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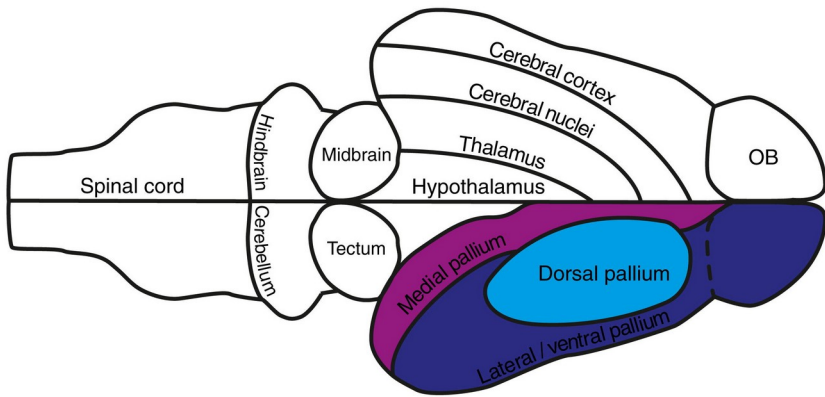
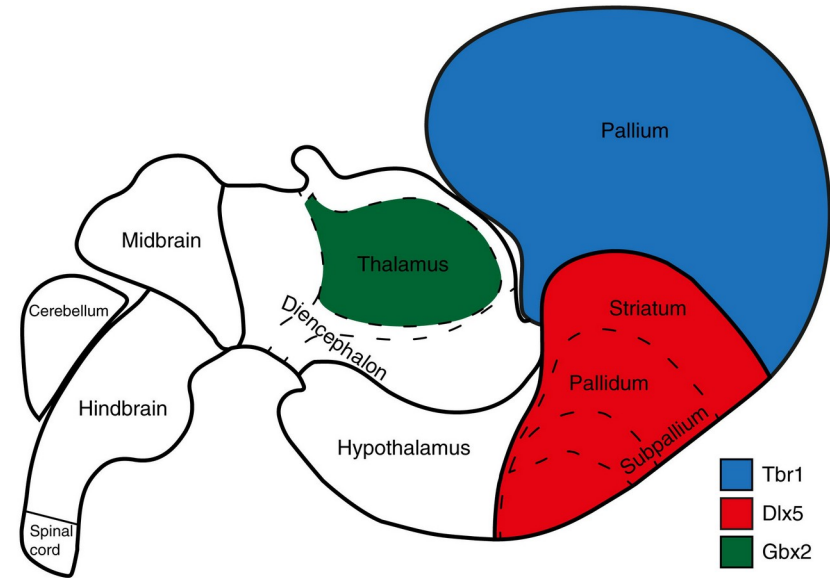
# Memory and Simulation in the Brain

## Integration: The Cortico-Hippocampal-Basal Ganglia Loop

- Cortex predicts.
- Hippocampus imagines.
- Basal ganglia choose.
- Together, they make thought actionable.



# Memory and Simulation in the Brain



Current Biology

## From Reptile Brain to Mammal Brain: Expanding the Predictive Circuit?

- Mammals didn't invent prediction.
- They expanded it—adding memory, imagination, and control.
- The neocortex scaled up an ancient loop.

# Memory and Simulation in the Brain

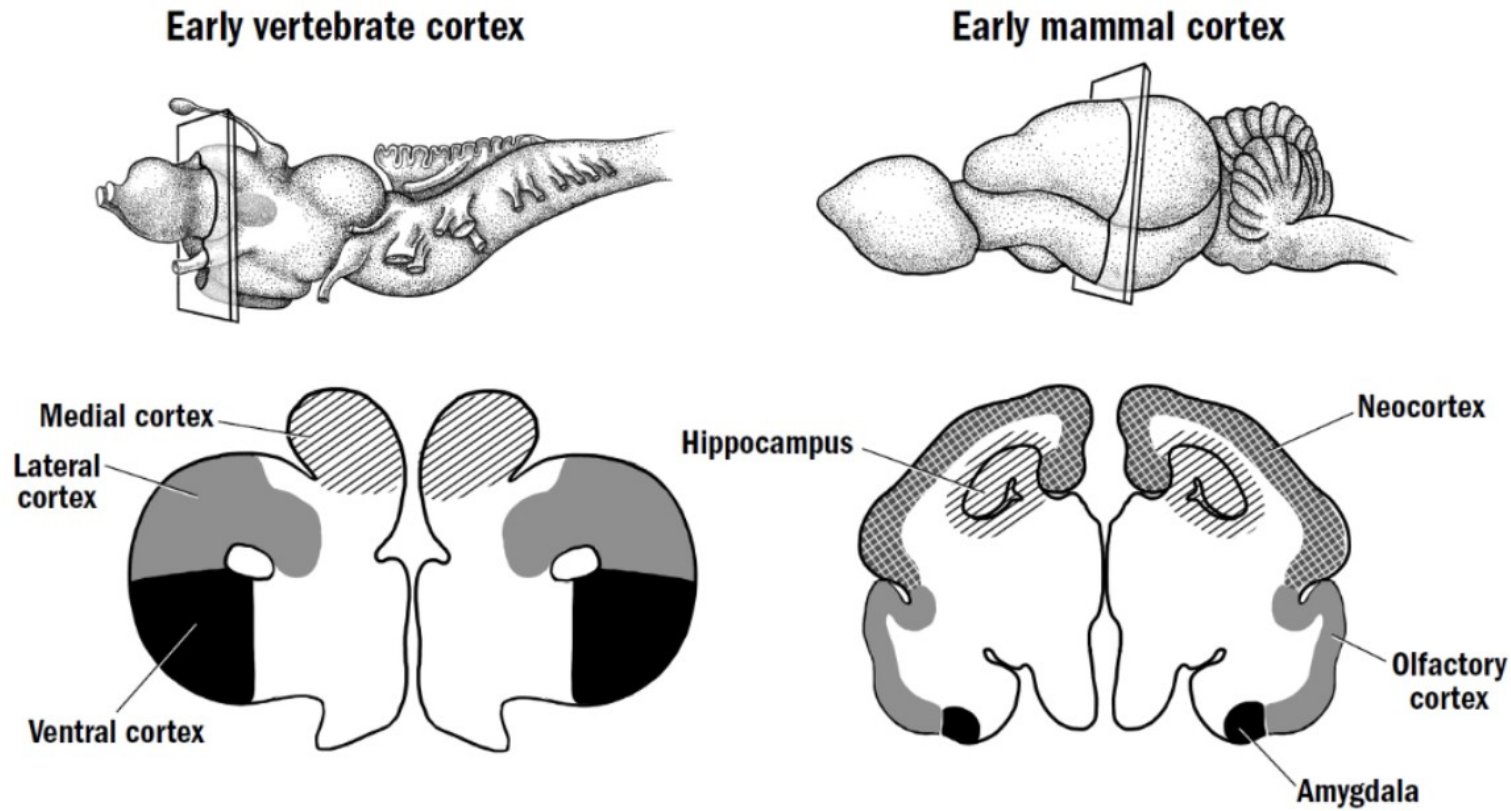


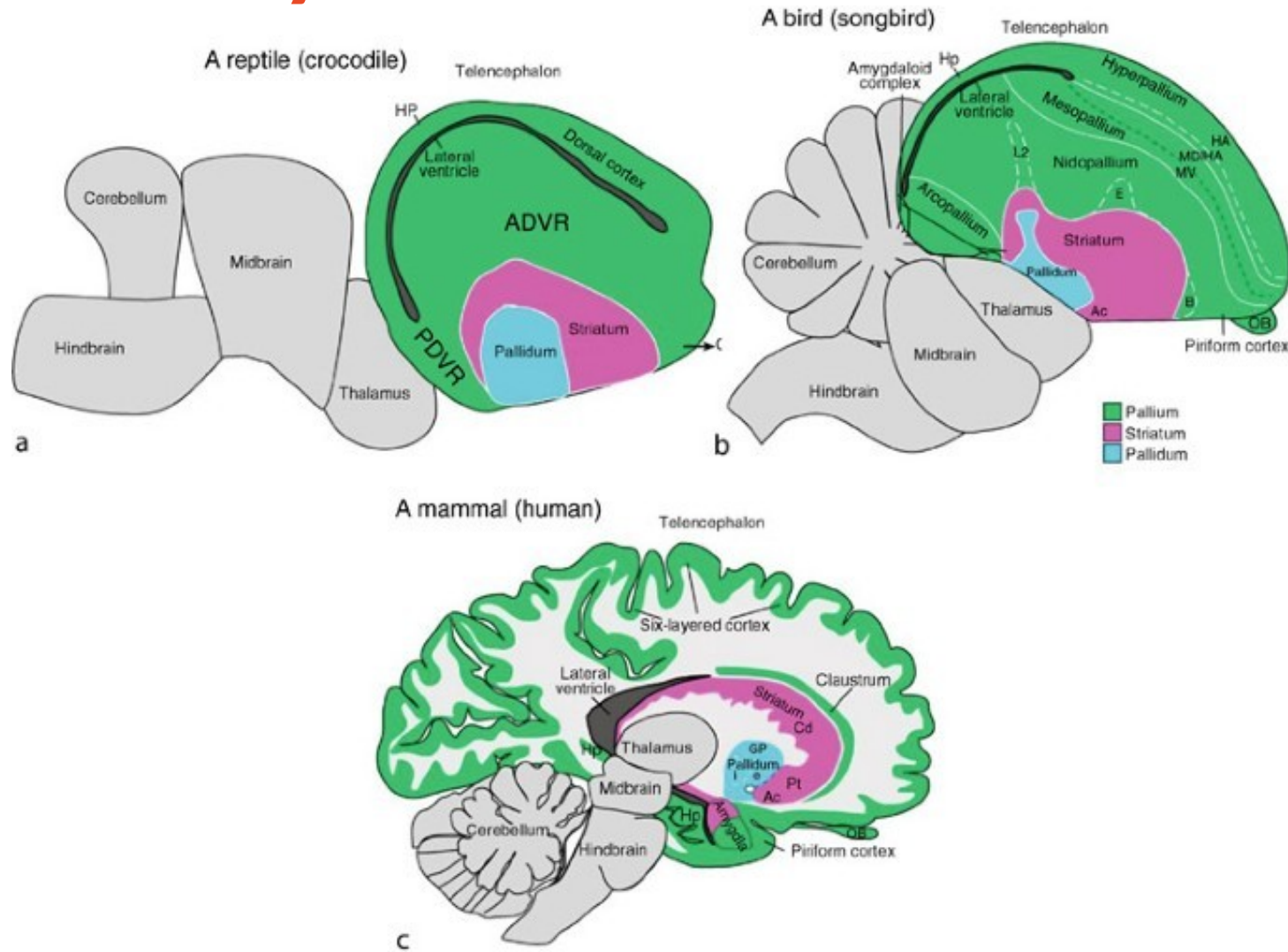
Figure 10.3: How the cortex changed in the transition from early vertebrates to early mammals

Original art by Rebecca Gelernter

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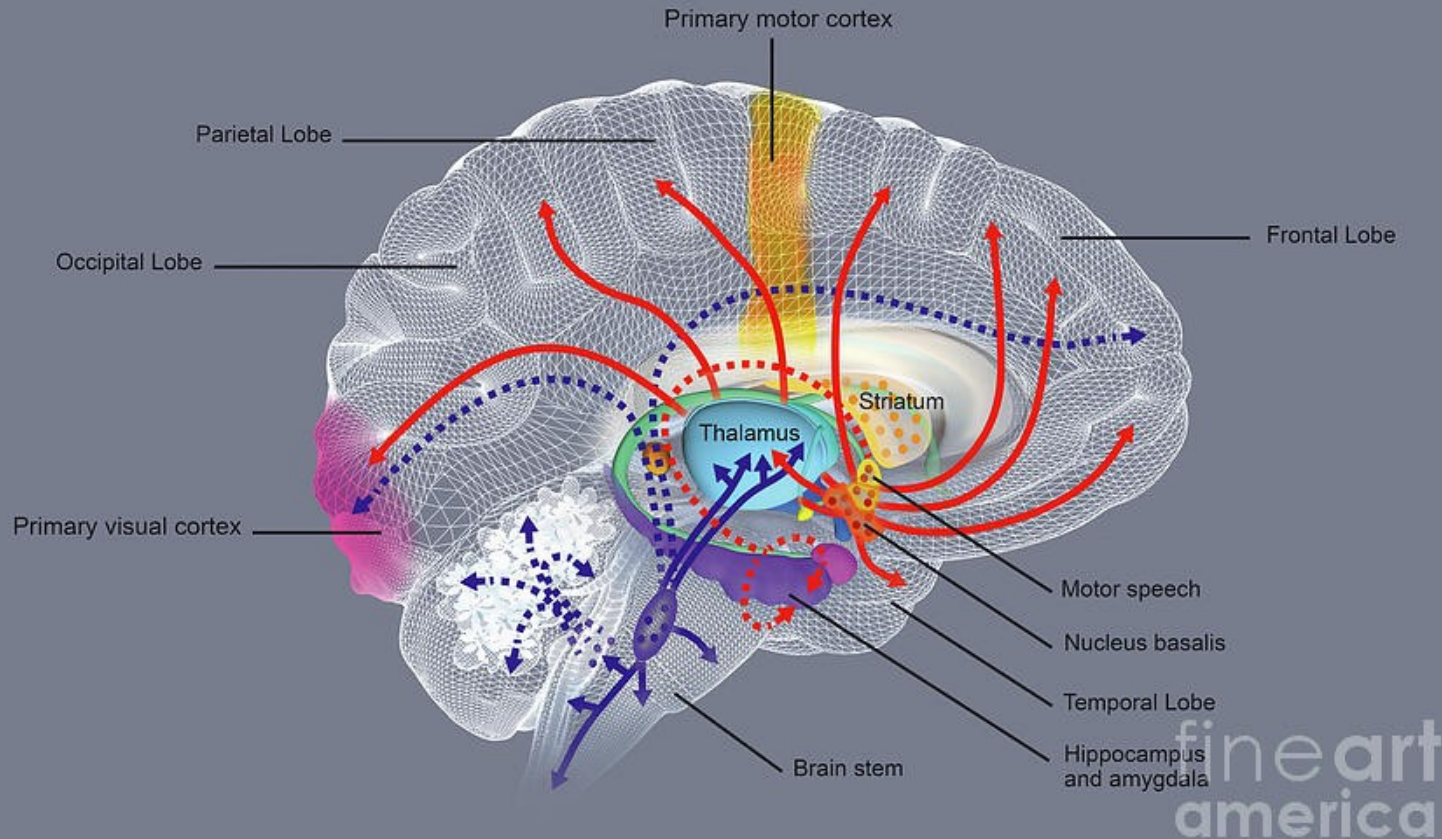
# Memory and Simulation in the Brain



## Birds and Convergent Circuits: Parallel Solutions to Prediction

- Birds lack a neocortex—but not imagination.
- The avian pallium evolved its own predictive circuits.
- Different anatomy, same computation.

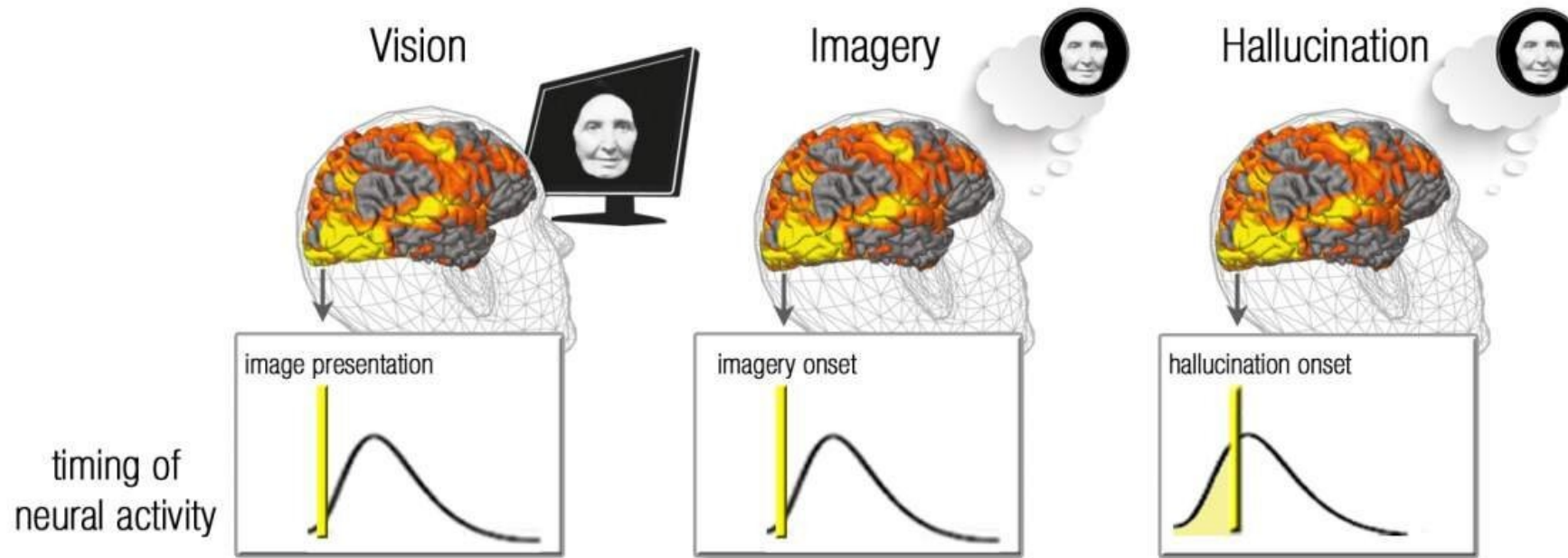
# Memory and Simulation in the Brain



## Dreaming and Predictive Maintenance

- Dreaming recalibrates the generative model.
- REM sleep replays, recombines, and repairs predictions.
- The brain updates itself offline.

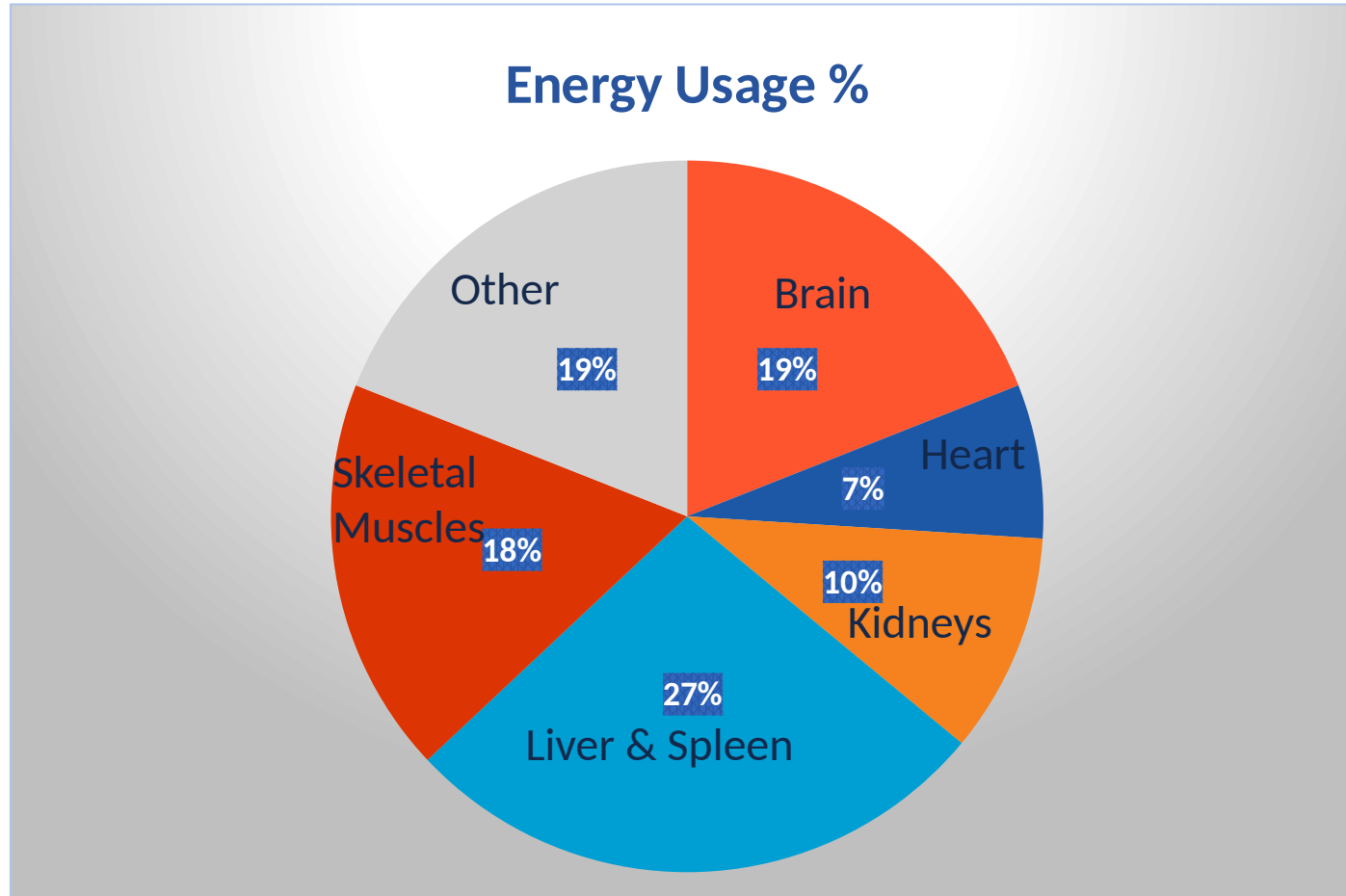
# Memory and Simulation in the Brain



## When Prediction Fails: Hallucination, Delusion, and Sensory Overload

- Too much prediction → hallucination.
- Too much input → chaos.
- Perception depends on balance.

# Memory and Simulation in the Brain



## Energy and Efficiency: The Cost of Prediction

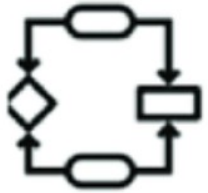
- Prediction saves energy—
- but maintaining the model is expensive.
- Brains trade computation for surprise reduction.

# Memory and Simulation in the Brain



## Computation

*What does the system do?*



## Algorithm

*How does the system do it?*



## Implementation

*How is the system realized?*

## Summary

- Prediction is the brain's core computation.
- Memory and imagination are its extensions.

## Next Time

- Social Cognition: How minds learned to model each other.